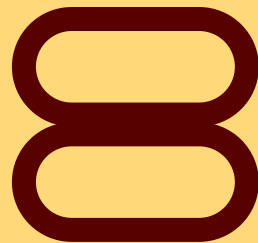
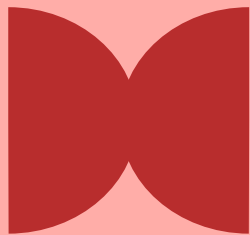




AI Governance



CS 120: Introduction to AI Safety
Naomi Solomon
November 11, 2025

When you hear AI governance,
what comes to mind?

How is it different than AI
safety?



AI Governance

institutions, laws, policies, norms, standards, and enforcement mechanisms shaping how AI is developed and deployed

AI Safety

technical solutions to ensure that AI systems operate safely and reliably



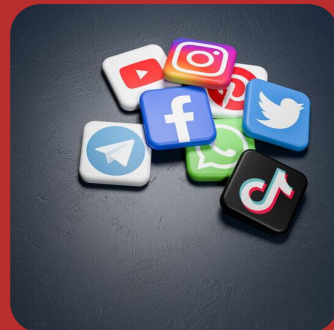
Why is AI hard to govern?



Nuclear
Technology



Genetic
Engineering



Internet/Social
Media

Why is AI hard to govern?

Nuclear Technology

Analogies: Dual-use, existential risk

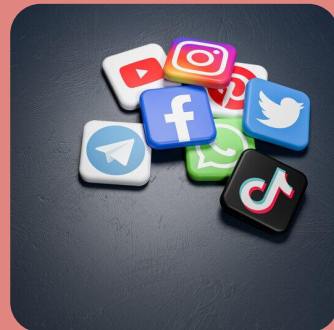
Disanalogies: Public vs Private Innovation; Physical vs Digital Assets



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Why is AI hard to govern?

Genetic Engineering

Analogies: Private development + public regulation, high commercial upsides

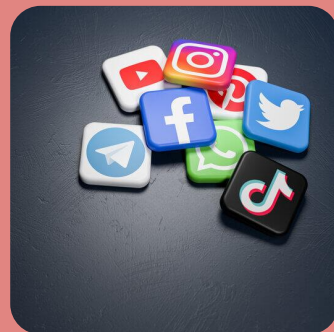
Disanalogies: Physical vs Digital Assets; Consensus on safety norms



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Technology



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Why is AI hard to govern?

Internet/Social Media

Analogies: Private innovation, digital asset

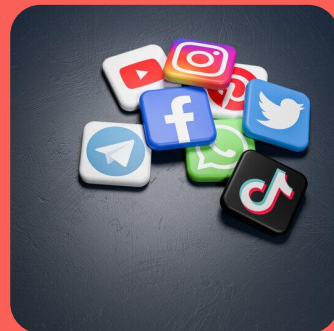
Disanalogies: Internet lacked an initial risk assessment, Open vs proprietary



Nuclear
Technology



Genetic
Engineering



Internet/Social
Media

Why is AI hard to govern?

Digital, diffuse, and fast-moving

Privately led, globally fragmented

No consensus on risk or norms

Anti-regulatory, anti-institutional culture

Dual-use

Emergent and unpredictable capabilities

b. Implementing Minimum Risk Management Practices for High-Impact AI

Agencies must implement the following minimum risk management practices for high-impact AI use cases:

- i. Conduct Pre-Deployment Testing. Agencies must develop pre-deployment testing and prepare risk mitigation plans that reflect expected real-world outcomes and identify expected benefits to the AI use. In conducting pre-deployment testing, if an agency does not have access to the underlying AI source code, models, or data, the agency must use alternative test methodologies, such as querying the AI service and observing the outputs or providing evaluation data to the vendor and obtaining results.

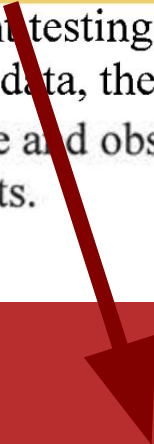
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How does an agency determine what is a robust alternative test methodology?



Which pre-deployment testing strategies must an agency use to ensure adequate safety measures have been met?

Gaps in AI Governance

Lack of standardized evaluation and red-teaming

No enforcement mechanism for safety standards

Limited interpretability and auditability

No consensus on safety metrics

Limited institutional capacity for technical expertise

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NIST AI Risk Management Framework

NIST AI RMF – Part 1. Trusted AI Systems and Organizational Risks

Part 1 of the NIST AI RMF focuses on defining what makes an AI system “trusted.” It lays out key principles like reliability, transparency, fairness, accountability, and security, and also introduces common AI risks such as:

- **Bias:** AI Algorithms reflecting unintentional discrimination
- **Privacy violations:** AI pipelines mishandling sensitive data
- **Security gaps:** AI systems’ vulnerabilities that attackers can exploit
- ...and many others.

NIST AI Risk Management Framework

Core Function	What it helps you do	Why it matters
Govern	Define governance structures, assign roles, and outline responsibilities for managing AI risks	Helps align AI systems with standards, regulations, and organizational values
Map	Identify and assess risks throughout the AI lifecycle	Fosters proactive identification of risks to promote AI security, and ensures AI aligns with governance practices
Measure	Quantify and assess the performance, effectiveness, and risks of AI systems	Ensures that AI remains stable, efficient, and compliant over time
Manage	Develop strategies for mitigating risks and ensuring AI systems remain compliant and secure	Facilitates continuous monitoring, auditing, and improvement to minimize risk exposure

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So, where to go from here?

ANDREESSEN
HOROWITZ

Regulate AI Use, Not AI Development

Matt Perault

<https://a16z.com/regulate-ai-use-not-ai-development/>

Marc Andreessen, founder of the Silicon Valley venture capital firm Andreessen Horowitz (a16z), has published a "manifesto" in which he lists "enemies" of technological progress and artificial intelligence specifically. Impeding the development of AI in any way, he argues, "is a form of murder."

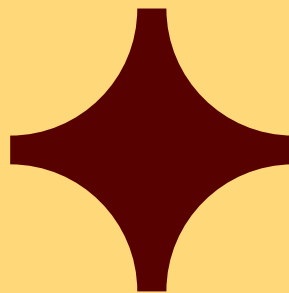
<https://www.404media.co/marc-andreessen-manifesto-says-ai-regulation-is-a-form-of-murder/>

Prior strategies of governing emerging technology cannot be copy and pasted onto the field of AI

Dynamic forms of governance need to be explored: how does a slow moving governance body keep up with Silicon Valley?

Continued investment into AI safety research and places for technical expertise in governance bodies

Takeaways





Stanford University
McCoy Family Center
for Ethics in Society



Stanford University
Human-Centered
Artificial Intelligence

Tech Ethics & Policy Summer Fellowships



December 1, 2025

Round 1 applications due